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FORREST BICKER

Mathematics + Computer Science

in forrestbicker

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EDUCATION

Harvey Mudd College

B.S. Mathematics and Computer Science

- 4.0 Major GPA, ICPC, ML Research at {AMISTAD Lab, Backgammon Lab}

Claremont, CA

Aug 2021 - May 2025

Hunter College High School

New York, NY

EXPERIENCE



Arrowstreet Capital

Quantitative Researcher

Boston, MA

Jul 2025 - Present



DE Shaw & Co.

Quant Intern

New York, NY

Jun - Aug 2024

- Researched enhancements to portfolio optimization reward model.
- Used numerical methods to improve quadratic program solver efficiency.



Amazon Web Services

Software Engineer Intern

Seattle, WA

Jul - Sep 2022

- Robustified AWS Kubernetes provisioning and recovery systems to orchestrate 10,000-node ultra-scale GPU clusters.

RESEARCH & PROJECTS

Backgammon Research Group

Backgammon Research under Dr. Arthur Benjamin

Claremont, CA

Nov 2021 - May 2022

- Designed quantitative strategies for tournament backgammon play under world champion Dr. Arthur Benjamin.
- Developed novel betting strategy that outperforms state-of-the-art methods and proved successful in championship play.

AMISTAD Lab

Machine Learning Research under Dr. George Montañez

Claremont, CA

May - Sep 2022

- Proved theoretical bounds on the Bias-Expressivity tradeoff for continuous learning problems and fuzzy membership.
- Published “From Targets to Rewards: Continuous Target Sets in the Algorithmic Search Framework” in ICAART 2024

Reinforcement Learning Research

Self-Play RL Agents

Claremont, CA

Sep 2023 - May 2024

- Built a reinforcement learning research system that placed top 3 globally out of 25,000 teams and 100,000 submitted algorithms in a machine learning competition.
- Notes: Convolutional transformer actor-critic architecture, auto-regressive action decoding, behavioral cloning on 1M expert games, Prioritized Fictitious Self Play league RL across 20 million episodes, parallelized PPO with concurrent rollouts across 72 workers and async GPU learner, shallow MCTS for efficient inference time compute.

AWARDS

- 1st Place Paper at Citadel Global Data Open Championship (\$100,000 prize)
- 1st Place at IMC Prosperity Global Trading Challenge (\$25,000 prize, 1st of 13,500)
- Top 50 USA at ICPC North American Championship (International Collegiate Programming Contest NAC-NAPC)
- 1st Place (2x) at West Coast Datathon (\$20,000 prize)
- 1st Place at National Security Agency (NSA) Cybersecurity Hacklytics Challenge
- 3rd Place Algorithm at Worldwide Correlation One Terminal (\$2,500 prize, 3rd of 25,000)

SKILLS

🐍 Python, 🔥 Pytorch, 📦 NumPy, 📊 Pandas, 🧠 Statsmodels, 🏹 CVXPY